



CAPSTONE PROJECT
DSA0404 - FUNDAMENTALS OF DATA SCIENCE

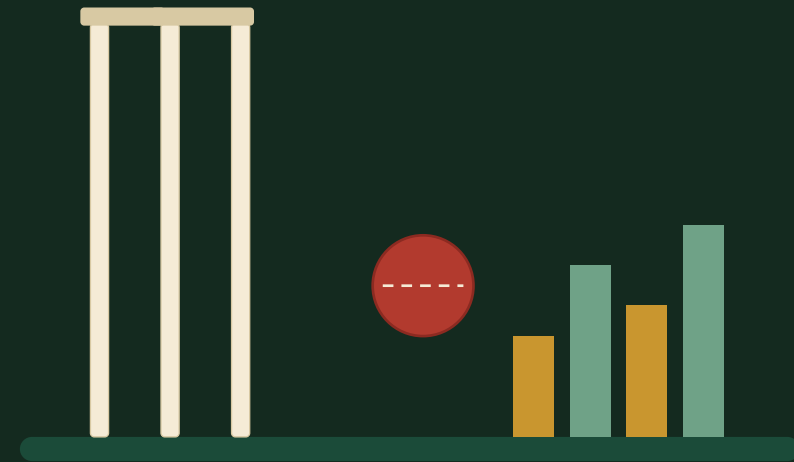


Cricket Research Lab

*A Data Science Platform for Investigating Historical
Patterns in International Test and One-Day Cricket*

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OBJECTIVES

FROM RAW CRICKET DATA TO A UNIFIED RESEARCH WORKSPACE

	OBJECTIVE	OUTCOME
①	Build a unified research platform for historical international cricket data	Cricket Research Lab - a single analytical workspace
②	Design a reliable ETL pipeline for Cricsheet JSON	Structured pipeline converting raw JSON into tabular data
③	Extract player, team, and match-level statistics from raw deliveries	Batting, bowling & match statistics computed from delivery-level data
④	Generate clean, structured analytical datasets	Consistent, query-ready datasets across matches & players
⑤	Develop an interactive analytics interface	Four dedicated views: Observatory, Dossier, Squad Matrix, Post-Mortem
⑥	Establish a reusable foundation for future cricket research	Structured analytical datasets ready for extended analysis and predictive modelling



WHY THIS PLATFORM?

THE MOTIVATION BEHIND CRICKET RESEARCH LAB



01

THE PROBLEM

- Historical cricket data exists as raw, nested match records.
- Direct analysis requires repeated data preparation and preprocessing.
- Existing workflows remain fragmented across scripts and datasets.

02

THE NEED

- Convert raw records into structured, reusable analytical data.
- Bring player, team and match analysis into one environment.
- Establish a consistent workflow for systematic historical research.

03

WHAT IT ENABLES

- Track player careers across formats, seasons and match history.
- Compare team form, performance and head-to-head records.
- Reconstruct the historical scorecards and innings progression.



PROBLEM STATEMENT

WHY A CENTRALIZED ANALYTICAL ENVIRONMENT IS NEEDED

MAIN CHALLENGE

Raw, unstructured historical match data cannot be directly analyzed - it must first be transformed into a structured, centralized form.

SPECIFIC ISSUES

- Nested ball-by-ball JSON cannot be queried directly
- Manual preprocessing repeated for every analysis
- No built-in charts or dashboards for trend exploration
- Player, team & tournament insights remain scattered
- Data preparation consumes most available analysis time

OUR SOLUTION

- ETL pipeline converts raw JSON into structured datasets
- Analytics engine derives player, team & match statistics
- Interactive Dash interface replaces manual scripting

Implemented as the Cricket Research Lab



LITERATURE REVIEW
PRIOR WORK IN CRICKET & SPORTS ANALYTICS



Sl. No.	Paper Title	Authors	Year	Techniques	Findings	Limitations
1	Cricket data analytics: forecasting T20 match winners through ML	Chakraborty et al.	2024	RF, SVM, XGBoost, LogReg	Random Forest reached 84.06% winner-prediction accuracy	Trained on team-level data only; no player granularity
2	Sport analytics for cricket game results using ML: an experimental study	Kapadia, Abdel-Jaber, Thabtah, Hadi	2019	Naive Bayes, RF, KNN, Model Trees	Random Forest outperformed others on IPL result prediction	Toss-based features gave weak predictive power
3	Cricket match analytics using the big data approach	Awan et al.	2021	Linear Regression, Spark ML	Spark ML improved scalability; 95% score-prediction accuracy	High infrastructure overhead for big-data tooling
4	Big data analytics framework for decision-making in sports performance	Mănescu, D.C.	2025	Gradient Boosting, MLP, ETL (Airflow)	Integrated ETL + analytics framework cut injury risk by 12%	Validated on synthetic, not live match data
5	A machine learning framework for sport result prediction	Bunker, Thabtah	2019	ML pipeline design (review + framework)	Proposes a reusable feature-engineering framework for sport ML	Framework not benchmarked on cricket specifically



EXISTING SYSTEM FRAGMENTED WORKFLOWS VS. A UNIFIED PLATFORM



EXISTING APPROACH: SCATTERED, AD-HOC WORKFLOWS

Raw JSON Files

One-off Parsing Script

Manual Stat Notes

Ad-hoc Spreadsheet

Unshared Local Charts

X no shared pipeline X no central dataset X nothing reusable

CRICKET RESEARCH LAB: UNIFIED ANALYTICAL PLATFORM

- 1 **Structured Data**
One persistent, queryable dataset spanning matches & players
- 2 **Visualization**
Built-in dashboards & charts for trend exploration
- 3 **Centralization**
Player, team & match insights in a single interface
- 4 **Reusability**
One pipeline serves every subsequent research question

THE CORE GAP - no centralized, structured environment for systematic historical investigation.



METHODOLOGY

RESEARCH PIPELINE & DEVELOPMENT APPROACH



TOOLS & FRAMEWORKS USED

Development

- Python
- Pandas
- Dash
- Plotly
- Dash Bootstrap Components

Data Source - Cricsheet

DEVELOPMENT APPROACH

Individual, iterative development - built incrementally from the ETL foundation toward a functioning analytical interface.

DATA COLLECTION

Ball-by-ball JSON for international Test & ODI matches, sourced from the open Cricsheet dataset.

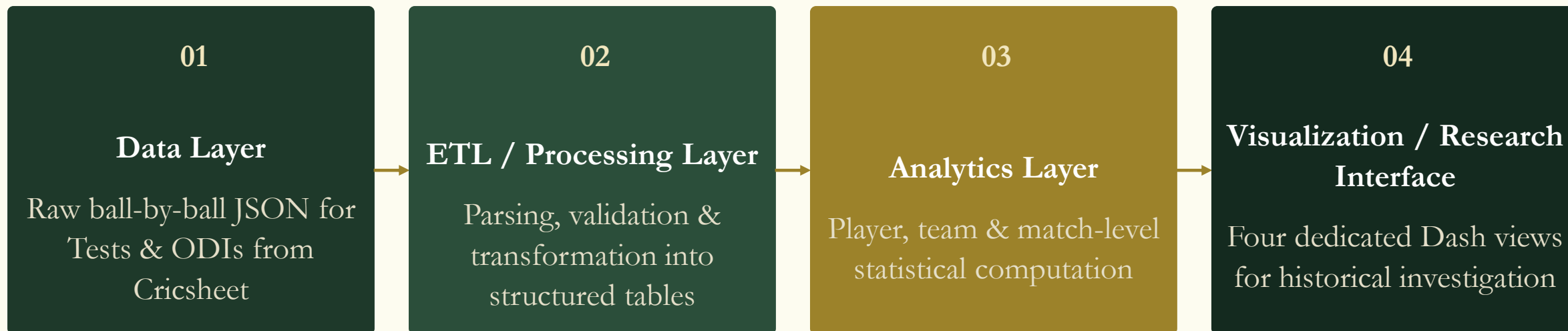
PROJECT PIPELINE





SYSTEM DESIGN

CRICKET RESEARCH LAB - THE COMPLETE ANALYTICAL ENVIRONMENT



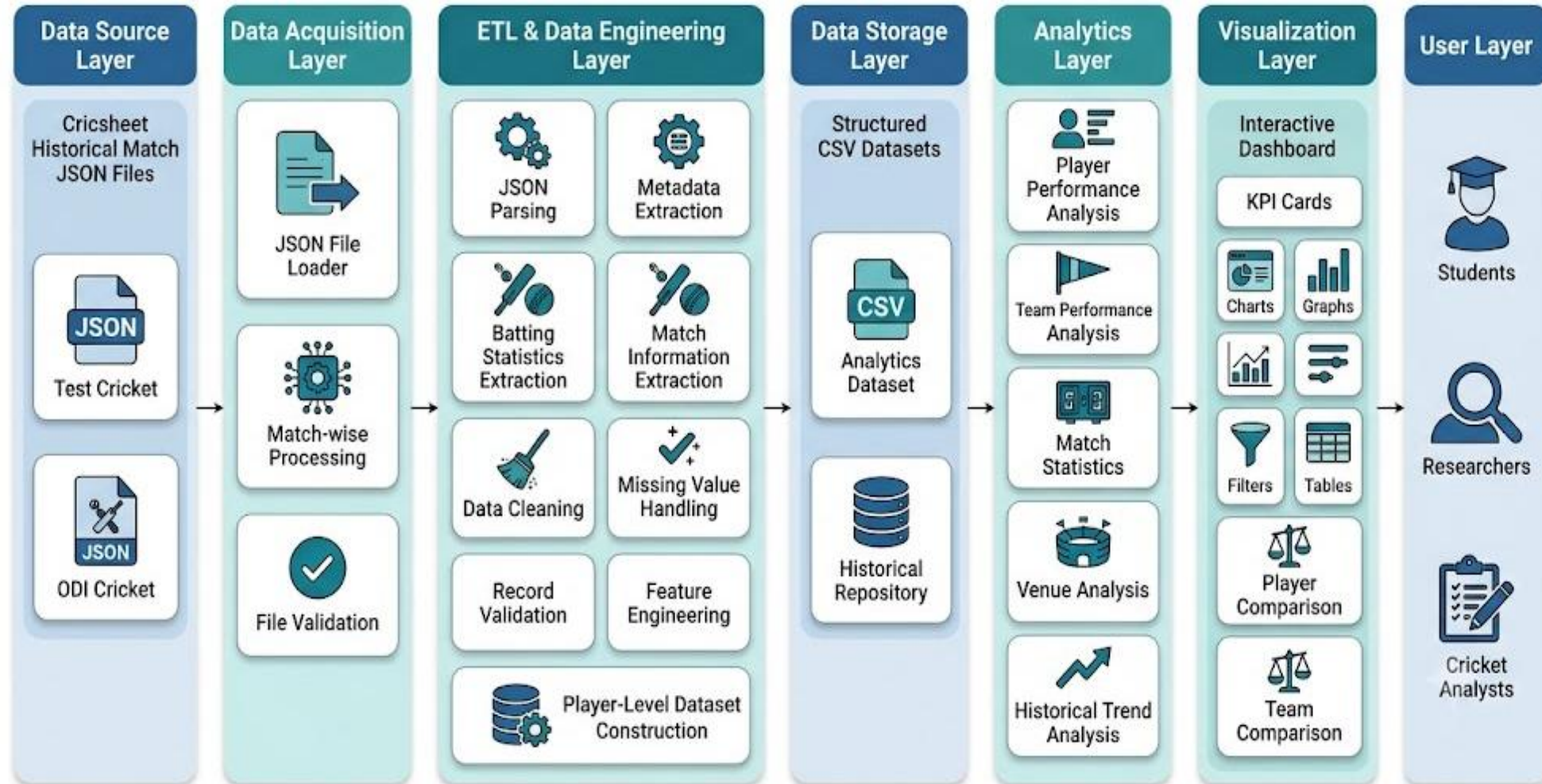
HOW THIS ADDRESSES THE CORE GAP

The Existing System analysis identified the core gap as the absence of a centralized, structured analytical environment. Cricket Research Lab closes this gap directly: the ETL layer removes manual preprocessing, the analytics layer standardizes statistics across players, teams & matches, and the Dash-based research interface replaces scattered, one-off scripts with a single, reusable workspace.



SYSTEM ARCHITECTURE

END-TO-END PLATFORM ARCHITECTURE

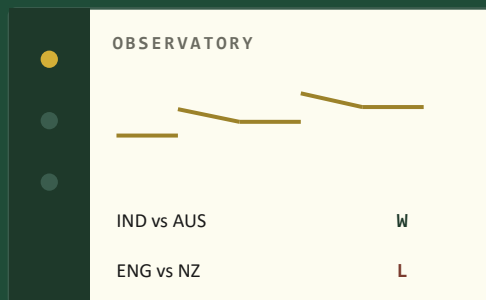




THE PLATFORM ANALYTICAL WORKSPACE



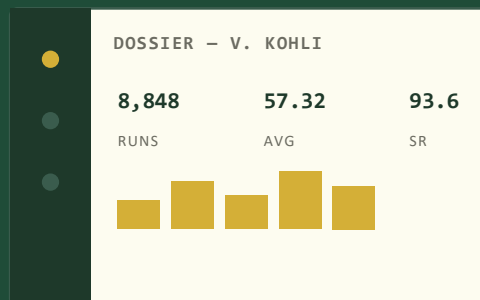
Four Views for Historical Cricket Investigation



01

THE OBSERVATORY

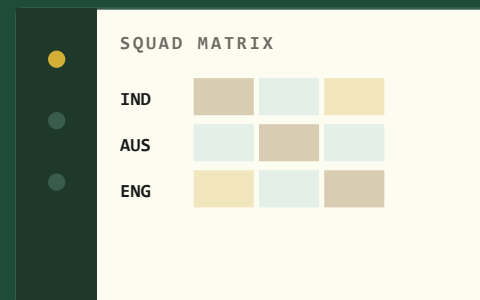
Global overview, recent fixtures, form & historical landscape.



02

THE DOSSIER

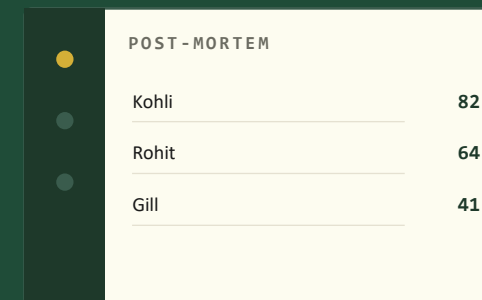
Individual player performance, career ledger, innings form & metric trajectory.



03

THE SQUAD MATRIX

Team form, roster composition & head-to-head analysis.



04

THE POST-MORTEM

Match scorecard, innings progression & partnership analysis.

Representative interface mockups in the Cricket Research Lab design language — illustrative, not literal screenshots.



RESULTS & DISCUSSION

DEMONSTRABLE OUTCOMES FROM THE PLATFORM



→ Structured Transformation

Raw, ball-by-ball Cricsheet JSON is successfully transformed into a structured, analysis-ready dataset.

→ Dedicated Analytical Views

Player, team & match-level information is explorable through four purpose-built research views.

→ Reduced Manual Inspection

The interface removes the need to manually inspect or re-parse raw JSON for routine questions.

→ Unified Research Environment

Multiple forms of cricket analysis - player, squad, and match are brought into one workspace.

→ Multi-Level Investigation

The system supports comparative (Squad Matrix), longitudinal (Dossier) & match-level (Post-Mortem) analysis.



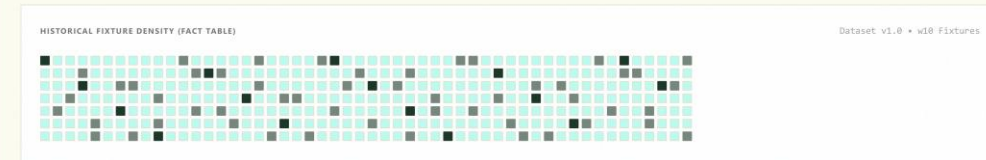
Lab Overview

Global landscape & recent fixture diagnostics

MATCHES 10 RUNS 10,412 WICKETS 321

RECENT FIXTURES TAPE			CRICHSHEET ACTIVE MATCHES
20 JAN 2017	New Zealand vs Bangladesh	MATCH CONCLUDED	POTM: TG SOUTHEE
12 JAN 2017	New Zealand vs Bangladesh	MATCH CONCLUDED	POTM: TWM LATHAM
03 JAN 2017	Australia vs Pakistan	MATCH CONCLUDED	POTM: DA WARNER
26 DEC 2016	Australia vs Pakistan	MATCH CONCLUDED	POTM: SPD SMITH
15 DEC 2016	Australia vs Pakistan	MATCH CONCLUDED	POTM: ASAD SHAFIQ
25 NOV 2016	New Zealand vs Pakistan	MATCH CONCLUDED	POTM: TG SOUTHEE
24 NOV 2016	Australia vs South Africa	MATCH CONCLUDED	POTM: UT KHAWAJA

TOP BATSMEN FORM INDEX	
01	SPD Smith 653 Runs (SR: 62.61)
02	DA Warner 592 Runs (SR: 93.38)
03	UT Khawaja 581 Runs (SR: 54.1)
04	Azhar Ali 511 Runs (SR: 48.27)
05	PSP Handscomb 399 Runs (SR: 59.73)





CONCLUSION & FUTURE SCOPE



CURRENT CONTRIBUTION

- Cricket Research Lab delivers a working analytical platform for historical cricket data
- ETL pipeline reliably structures raw Cricsheet JSON into analysis-ready tables
- Four analytical views operationalize the project's original objectives
- Replaces fragmented, script-based analysis with one unified workspace
- Establishes a reusable foundation for further research

FUTURE SCOPE

- Broaden dataset coverage across formats & tournaments
- Add further analytical indicators & comparative metrics
- Improve filtering, search & cross-view navigation
- Explore future predictive / advanced analytics
- Quantitative benchmarking (accuracy, performance metrics, user studies)



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- S. Chakraborty, A. Mondal, A. Bhattacharjee, A. Mallick, R. Santra, S. Maity, and L. Dey, “Cricket data analytics: Forecasting T20 match winners through machine learning,” *Int. J. Knowl.-Based Intell. Eng. Syst.*, vol. 28, no. 1, pp. 1–14, 2024.
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THANK YOU!